Computer Architecture
CS 622 / EE 520
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Lecture 14

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Topics

- Branch Prediction and Speculation
- Some solved Examples
  - Loops / Unrolling
  - VLIW
  - Speculative Execution

Quiz 3
On Monday

Midterm
Next Week

Assignment
Due Monday

Further Readings

Reading Notes

Readings:
Chapter 3 of course textbook

Exploiting ILP Using Dynamic Scheduling, Multiple Issue, and Speculation

ILP Using Dynamic Schedule, Multiple Issue and Speculation – Put all together

So far we have seen how the individual mechanisms of dynamic scheduling, multiple issue, and speculation work. In this section, we put all three together, which yields a microarchitecture quite similar to those in modern microprocessors. For simplicity we consider only an issue rate of two instructions per clock, but the concepts are no different from modern processors that issue three or more instructions per clock.

Let's assume we want to extend Tomasulo's algorithm to support multiple-issue superscalar pipeline with separate integer, load/store, and floating-point units (both FP multiply and FP add), each of which can initiate an operation on every clock. We do not want to issue instructions to the reservation stations out of order because this could lead to a violation of the program semantics. To gain the full advantage of dynamic scheduling, we will allow the pipeline to issue any combination of two instructions in a clock, using the scheduling hardware to actually assign operations to the integer and floating-point unit. Because the interaction of the integer and floating-point instructions is crucial, we also extend Tomasulo's scheme to deal with both the integer and floating-point functional units and registers, as well as incorporating speculative execution. As Figure 3.21 shows, the basic organization is similar to that of a processor with speculation with one issue per clock, except that the issue and completion logic must be enhanced to allow multiple instructions to be processed per clock.

Dealing with Multiple Instruction Issue per clock

Issuing multiple instructions per clock in a dynamically scheduled processor (with or without speculation) is very complex for the simple reason that the multiple instructions may depend on one another. Because of this, the tables must be updated for the instructions in parallel; otherwise, the tables will be incorrect or the dependence may be lost.

Two different approaches have been used to issue multiple instructions per clock in a dynamically scheduled processor, and both rely on the observation that the key is assigning a reservation station and updating the pipeline control tables.

1

One approach is to run this step in half a clock cycle so that two instructions can be processed in one clock cycle; this approach cannot be easily extended to handle four instructions per clock, unfortunately.

2

A second alternative is to build the logic necessary to handle two or more instructions at once, including any possible dependences between the instructions. Modern superscalar processors that issue four or more instructions per clock may include both approaches: They both pipeline and widen the issue logic. A key observation is that we cannot simply pipeline away the problem. By making instruction issues take multiple clocks because new instructions are issuing every clock cycle, we must be able to assign the reservation station and to update the pipeline tables so that a dependent instruction issuing on the next clock can use the updated information.

Multiple Issue Processor with Speculation

Superscalar

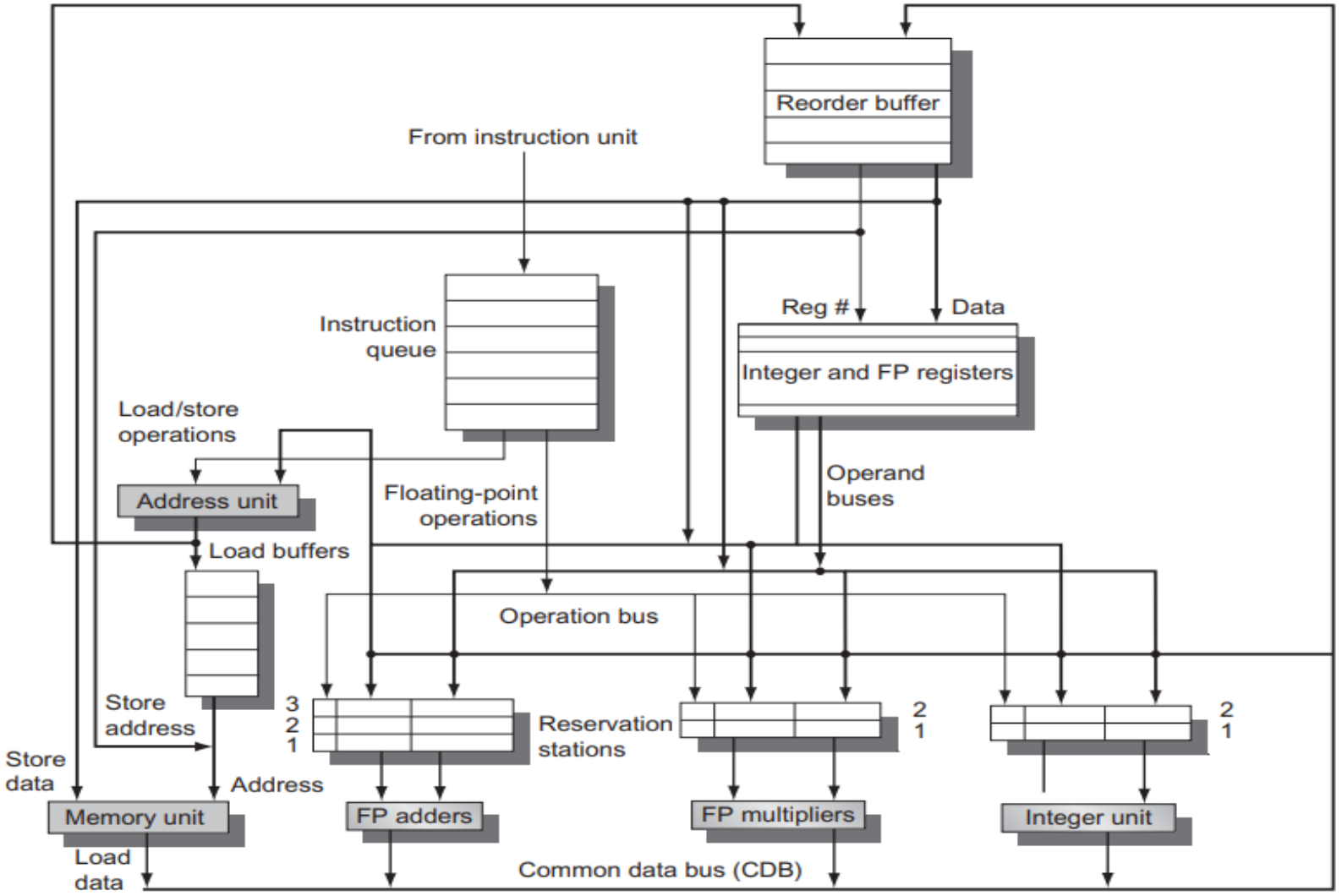


Figure 3.21 The basic organization of a multiple issue processor with speculation. In this case, the organization could allow a FP multiply, FP add, integer, and load/store to all issues simultaneously (assuming one issue per clock per functional unit). Note that several datapaths must be widened to support multiple issues: the CDB, the operand buses, and, critically, the instruction issue logic, which is not shown in this figure. The last is a difficult problem, as we discuss in the text.

Strategy for Dynamically Scheduled Superscalar

to describe the basic strategy for updating the issue logic and the reservation tables in a dynamically scheduled superscalar with up to *n* issues per clock as follows:

1. Assign a reservation station and a reorder buffer for *every* instruction that *might* be issued in the next issue bundle. This assignment can be done before the instruction types are known simply by preallocating the reorder buffer entries sequentially to the instructions in the packet using *n* available reorder buffer entries and by ensuring that enough reservation stations are available to issue the whole bundle, independent of what it contains. By limiting the number of instructions of a given class (say, one FP, one integer, one load, one store), the necessary reservation stations can be preallocated. Should sufficient reservation stations not be available (such as when the next few instructions in the program are all of one instruction type), the bundle is broken, and only a subset of the instructions, in the original program order, is issued. The remainder of the instructions in the bundle can be placed in the next bundle for potential issue.
2. Analyze all the dependences among the instructions in the issue bundle.
3. If an instruction in the bundle depends on an earlier instruction in the bundle, use the assigned reorder buffer number to update the reservation table for the dependent instruction. Otherwise, use the existing reservation table and reorder buffer information to update the reservation table entries for the issuing instruction.

Of course, what makes the preceding very complicated is that it is all done in parallel in a single clock cycle!

- ISSUE Step is a major bottleneck
- In a modern superscalar, every possible combination of dependent instructions that is allowed to issue in the same clock cycle must be considered.
- Because the number of possibilities climbs as the square of the number of instructions that can be issued in a clock, the issue step is a likely bottleneck for attempts to go beyond four instructions per clock.

Commit Strategy - continued

Of course, what makes the preceding very complicated is that it is all done in parallel in a single clock cycle!

At the back-end of the pipeline, we must be able to complete and commit multiple instructions per clock. These steps are somewhat easier than the issue problems because multiple instructions that can actually commit in the same clock cycle must have already dealt with and resolved any dependences. As we will see, designers have figured out how to handle this complexity: The Intel i7, which we examine in Section 3.12, uses essentially the scheme we have described for speculative multiple issue, including a large number of reservation stations, a reorder buffer, and a load and store buffer that is also used to handle nonblocking cache misses.

From a performance viewpoint, we can show how the concepts fit together with an example.

Observations from the Example (following slides)

Discussed separately in solved examples

Figures 3.23 and 3.24 show the performance for a two-issue, dynamically scheduled processor, without and with speculation. In this case, where a branch can be a critical performance limiter, speculation helps significantly. The third branch in the speculative processor executes in clock cycle 13, whereas it executes in clock cycle 19 on the nonspeculative pipeline. Because the completion rate on the nonspeculative pipeline is falling behind the issue rate rapidly, the nonspeculative pipeline will stall when a few more iterations are issued. The performance of the nonspeculative processor could be improved by allowing load instructions to complete effective address calculation before a branch is decided, but unless speculative memory accesses are allowed, this improvement will gain only 1 clock per iteration.

This example clearly shows how speculation can be advantageous when there are data-dependent branches, which otherwise would limit performance. This advantage depends, however, on accurate branch prediction. Incorrect speculation does not improve performance; in fact, it typically harms performance and, as we shall see, dramatically lowers energy efficiency.

Example page 226

Example Consider the execution of the following loop, which increments each element of an integer array, on a two-issue processor, once without speculation and once with speculation:

```
Loop:  ld    x2,0(x1)      //x2=array element
       addi  x2,x2,1       //increment x2
       sd    x2,0(x1)      //store result
       addi  x1,x1,8       //increment pointer
       bne   x2,x3,Loop    //branch if not last
```

Assume that there are separate integer functional units for effective address calculation, for ALU operations, and for branch condition evaluation. Create a table for the first three iterations of this loop for both processors. Assume that up to two instructions of any type can commit per clock.

Solution: 2 Issue Processor Without Speculation

Dynamically Scheduled

Iteration number	Instructions	Issues at clock cycle number	Executes at clock cycle number	Memory access at clock cycle number	Write CDB at clock cycle number	Comment
1	ld x2,0(x1)	1	2	3	4	First issue
1	addi x2,x2,1	1	5		6	Wait for ld
1	sd x2,0(x1)	2	3	7		Wait for addi
1	addi x1,x1,8	2	3		4	Execute directly
1	bne x2,x3,Loop	3	7			Wait for addi
2	ld x2,0(x1)	4	8	9	10	Wait for bne
2	addi x2,x2,1	4	11		12	Wait for ld
2	sd x2,0(x1)	5	9	13		Wait for addi
2	addi x1,x1,8	5	8		9	Wait for bne
2	bne x2,x3,Loop	6	13			Wait for addi
3	ld x2,0(x1)	7	14	15	16	Wait for bne
3	addi x2,x2,1	7	17		18	Wait for ld
3	sd x2,0(x1)	8	15	19		Wait for addi
3	addi x1,x1,8	8	14		15	Wait for bne
3	bne x2,x3,Loop	9	19			Wait for addi

Figure 3.23 The time of issue, execution, and writing result for a dual-issue version of our pipeline *without* speculation. Note that the ld following the bne cannot start execution earlier because it must wait until the branch outcome is determined. This type of program, with data-dependent branches that cannot be resolved earlier, shows the strength of speculation. Separate functional units for address calculation, ALU operations, and branch-condition evaluation allow multiple instructions to execute in the same cycle. Figure 3.24 shows this example with speculation.

Solution: 2 Issue Proc with Speculation

Dynamically Scheduled

Iteration number	Instructions	Issues at clock number	Executes at clock number	Read access at clock number	Write CDB at clock number	Commits at clock number	Comment
1	ld x2,0(x1)	1	2	3	4	5	First issue
1	addi x2,x2,1	1	5		6	7	Wait for ld
1	sd x2,0(x1)	2	3			7	Wait for addi
1	addi x1,x1,8	2	3		4	8	Commit in order
1	bne x2,x3,Loop	3	7			8	Wait for addi
2	ld x2,0(x1)	4	5	6	7	9	No execute delay
2	addi x2,x2,1	4	8		9	10	Wait for ld
2	sd x2,0(x1)	5	6			10	Wait for addi
2	addi x1,x1,8	5	6		7	11	Commit in order
2	bne x2,x3,Loop	6	10			11	Wait for addi
3	ld x2,0(x1)	7	8	9	10	12	Earliest possible
3	addi x2,x2,1	7	11		12	13	Wait for ld
3	sd x2,0(x1)	8	9			13	Wait for addi
3	addi x1,x1,8	8	9		10	14	Executes earlier
3	bne x2,x3,Loop	9	13			14	Wait for addi

Figure 3.24 The time of issue, execution, and writing result for a dual-issue version of our pipeline *with* speculation. Note that the ld following the bne can start execution early because it is speculative.

3.9 Advanced Techniques for Instruction Delivery and Speculation

Dealing With Branches

1

Branch-Target Buffers

To reduce the branch penalty for our simple five-stage pipeline, as well as for deeper pipelines, we must know whether the as-yet-undecoded instruction is a branch and, if so, what the next program counter (PC) should be. If the instruction is a branch and we know what the next PC should be, we can have a branch penalty of zero. A branch-prediction cache that stores the predicted address for the next instruction after a branch is called a *branch-target buffer* or *branch-target cache.*

2

Branch Folding

One variation on the branch-target buffer is to store one or more *target instructions* instead of, or in addition to, the predicted *target address*. This variation has two potential advantages. First, it allows the branch-target buffer access to take longer than the time between successive instruction fetches, possibly allowing a larger branch-target buffer. Second, buffering the actual target instructions allows us to perform an optimization called *branch folding*. Branch folding can be used to obtain 0-cycle unconditional branches and sometimes 0-cycle conditional branches. As we will see, the Cortex A-53 uses a single-entry branch target cache that stores the predicted target instructions.

Procedure Return Address Buffer

Though procedure returns can be predicted with a branch-target buffer, the accuracy of such a prediction technique can be low if the procedure is called from multiple sites and the calls from one site are not clustered in time. For example, in SPEC CPU95, an aggressive branch predictor achieves an accuracy of less than 60% for such return branches. To overcome this problem, some designs use a small buffer of return addresses operating as a stack. This structure caches the most recent return addresses, pushing a return address on the stack at a call and popping one off at a return. If the cache is sufficiently large (i.e., as large as the maximum call depth), it will predict the returns perfectly. Figure 3.28 shows the performance of such a return buffer with 0–16 elements for a number of the SPEC CPU95 benchmarks. We will use a similar return predictor when we examine the studies of ILP in Section 3.10. Both the Intel Core processors and the AMD Phenom processors have return address predictors.

In large server applications, indirect jumps also occur for various function calls and control transfers. Predicting the targets of such branches is not as simple as in a procedure return. Some processors have opted to add specialized predictors for all indirect jumps, whereas others rely on a branch target buffer.

Although a simple predictor like gshare does a good job of predicting many conditional branches, it is not tailored to predicting loop branches, especially for long running loops. As we observed earlier, the Intel Core i7 920 used a specialized loop branch predictor. With the emergence of tagged hybrid predictors, which are as good at predicting loop branches, some recent designers have opted to put the resources into larger tagged hybrid predictors rather than a separate loop branch predictor.

Subroutine Return Stack - Diagram

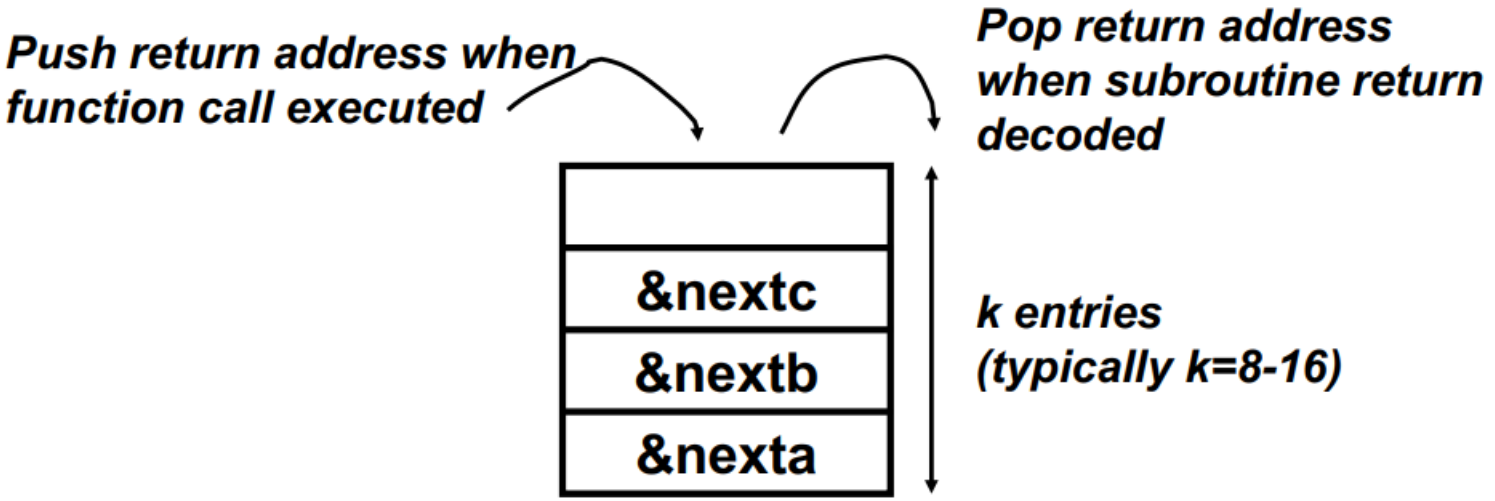
Subroutine Return Stack

Small structure to accelerate JR for subroutine returns, typically much more accurate than BTBs.

```
fa() { fb(); nexta: }

fb() { fc(); nextb: }

fc() { fd(); nextc: }
```



Integrated Instruction Fetch Unit

Instead, recent designs have used an integrated instruction fetch unit that integrates several functions:

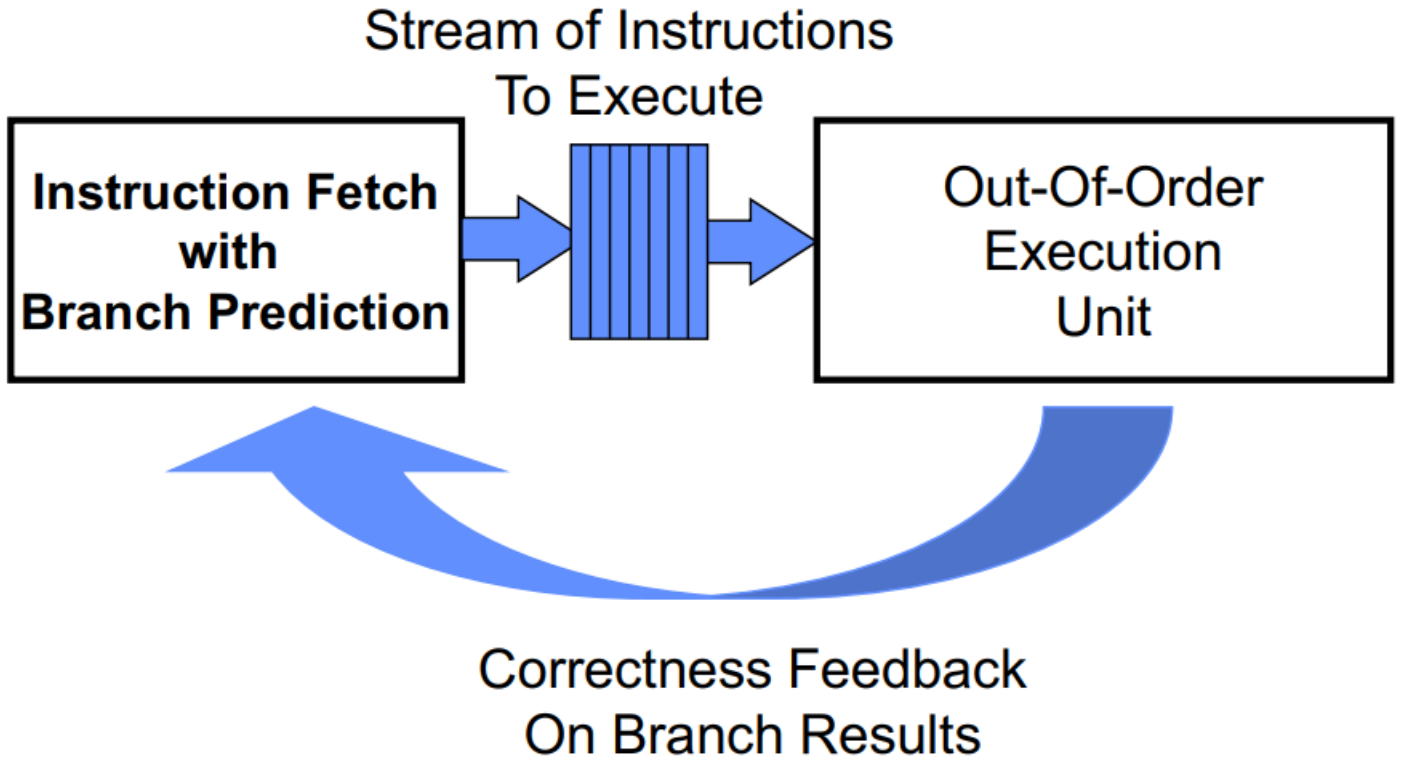
1. *Integrated branch prediction*—The branch predictor becomes part of the instruction fetch unit and is constantly predicting branches, so as to drive the fetch pipeline.
2. *Instruction prefetch*—To deliver multiple instructions per clock, the instruction fetch unit will likely need to fetch ahead. The unit autonomously manages the prefetching of instructions (see Chapter 2 for a discussion of techniques for doing this), integrating it with branch prediction.
3. *Instruction memory access and buffering*—When fetching multiple instructions per cycle, a variety of complexities are encountered, including the difficulty that fetching multiple instructions may require accessing multiple cache lines. The instruction fetch unit encapsulates this complexity, using prefetch to try to hide the cost of crossing cache blocks. The instruction fetch unit also provides buffering, essentially acting as an on-demand unit to provide instructions to the issue stage as needed and in the quantity needed.

Virtually all high-end processors now use a separate instruction fetch unit connected to the rest of the pipeline by a buffer containing pending instructions.

Independent Fetch Unit for Prefetching

Independent "Fetch" unit

- **Instruction fetch decoupled from execution**
  - **Instruction Buffer in-between**
- **Often issue logic (+ rename) included with Fetch**



Issuing Bundle of Instructions

Both register renaming and reorder buffers continue to be used in high-end processors, which now feature the ability to have as many as 100 or more instructions (including loads and stores waiting on the cache) in flight. Whether renaming or a reorder buffer is used, the key complexity bottleneck for a dynamically scheduled superscalar remains issuing bundles of instructions with dependences within the bundle. In particular, dependent instructions in an issue bundle must be issued with the assigned virtual registers of the instructions on which they depend. A strategy for instruction issue with register renaming similar to that used for multiple issue with reorder buffers (see page 205) can be deployed, as follows:

1. The issue logic reserves enough physical registers for the entire issue bundle (say, four registers for a four-instruction bundle with at most one register result per instruction).
2. The issue logic determines what dependences exist within the bundle. If a dependence does not exist within the bundle, the register renaming structure is used to determine the physical register that holds, or will hold, the result on which instruction depends. When no dependence exists within the bundle, the result is from an earlier issue bundle, and the register renaming table will have the correct register number.
3. If an instruction depends on an instruction that is earlier in the bundle, then the pre-reserved physical register in which the result will be placed is used to update the information for the issuing instruction.

Challenge of More Instruction Issues per Clock

Figure 3.29 shows a six-instruction code sequence and what the issue step must do. Remember that this must all occur in a single clock cycle, if the processor is to maintain a peak rate of six issues per clock! All the dependences must be detected, the physical registers must be assigned, and the instructions must be rewritten using the physical register numbers: in one clock. This example makes it clear why issue rates have grown from 3–4 to only 4–8 in the past 20 years. The complexity of the analysis required during the issue cycle grows as the square of the issue width, and a new processor is typically targeted to have a higher clock rate than in the last generation! Because register renaming and the reorder buffer approaches are duals, the same complexities arise independent of the implementation scheme.

How much to Speculate?

One of the significant advantages of speculation is its ability to uncover events that would otherwise stall the pipeline early, such as cache misses. This potential advantage, however, comes with a significant potential disadvantage. Speculation is not free. It takes time and energy, and the recovery of incorrect speculation further reduces performance. In addition, to support the higher instruction execution rate needed to benefit from speculation, the processor must have additional resources, which take silicon area and power. Finally, if speculation causes an exceptional event to occur, such as a cache or translation lookaside buffer (TLB) miss, the potential for significant performance loss increases, if that event would not have occurred without speculation.

To maintain most of the advantage while minimizing the disadvantages, most pipelines with speculation will allow only low-cost exceptional events (such as a first-level cache miss) to be handled in speculative mode. If an expensive exceptional event occurs, such as a second-level cache miss or a TLB miss, the processor will wait until the instruction causing the event is no longer speculative before handling the event. Although this may slightly degrade the performance of some programs, it avoids significant performance losses in others, especially those that suffer from a high frequency of such events coupled with less-than-excellent branch prediction.

In the 1990s the potential downsides of speculation were less obvious. As processors have evolved, the real costs of speculation have become more apparent, and the limitations of wider issue and speculation have been obvious. We return to this issue shortly.

Speculating through Multiple Branches

In the examples we have considered in this chapter, it has been possible to resolve a branch before having to speculate on another. Three different situations can benefit from speculating on multiple branches simultaneously: (1) a very high branch frequency, (2) significant clustering of branches, and (3) long delays in functional units. In the first two cases, achieving high performance may mean that multiple branches are speculated, and it may even mean handling more than one branch per clock. Database programs and other less structured integer computations, often exhibit these properties, making speculation on multiple branches important. Likewise, long delays in functional units can raise the importance of speculating on multiple branches as a way to avoid stalls from the longer pipeline delays.

Speculating on multiple branches slightly complicates the process of speculation recovery but is straightforward otherwise. As of 2017, no processor has yet combined full speculation with resolving multiple branches per cycle, and it is unlikely that the costs of doing so would be justified in terms of performance versus complexity and power.

Speculation and Energy Efficiency

What is the impact of speculation on energy efficiency? At first glance, one might argue that using speculation always decreases energy efficiency because whenever speculation is wrong, it consumes excess energy in two ways:

1. Instructions that are speculated and whose results are not needed generate excess work for the processor, wasting energy.
2. Undoing the speculation and restoring the state of the processor to continue execution at the appropriate address consumes additional energy that would not be needed without speculation.

Certainly, speculation will raise the power consumption, and if we could control speculation, it would be possible to measure the cost (or at least the dynamic power cost). But, if speculation lowers the execution time by more than it increases the average power consumption, then the total energy consumed may be less.

Thus, to understand the impact of speculation on energy efficiency, we need to look at how often speculation is leading to unnecessary work. If a significant number of unneeded instructions is executed, it is unlikely that speculation will improve running time by a comparable amount. Figure 3.30 shows the fraction of instructions that are executed from misspeculation for a subset of the SPEC2000 benchmarks using a sophisticated branch predictor. As we can see, this fraction of executed misspeculated instructions is small in scientific code and significant (about 30% on average) in integer code. Thus it is unlikely that speculation is energy-efficient for integer applications, and the end of Dennard scaling makes imperfect speculation more problematic. Designers could avoid speculation, try to reduce the misspeculation, or think about new approaches, such as only speculating on branches that are known to be highly predictable.

Explicit Register Renaming Process

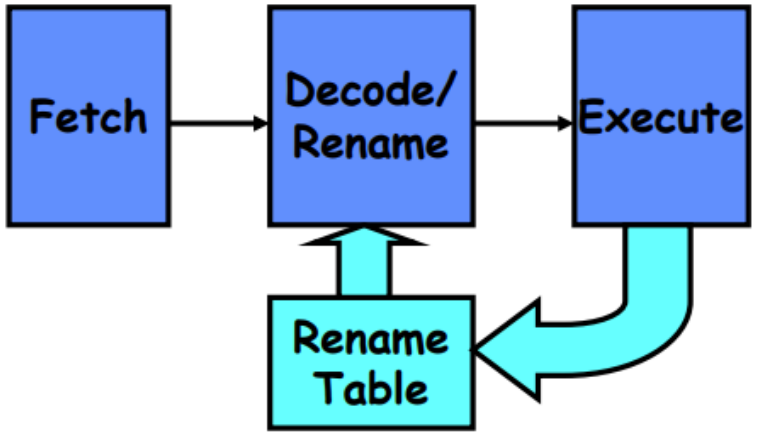
Explicit Register Renaming

- Tomasulo provides *Implicit Register Renaming*
  - User registers renamed to reservation station tags
- Explicit Register Renaming:
  - Use *physical* register file that is larger than number of registers specified by ISA
- Keep a translation table:
  - ISA register => physical register mapping
  - When register is written, replace table entry with new register from freelist.
  - Physical register becomes free when not being used by any instructions in progress.
- Pipeline can be exactly like "standard" DLX pipeline
  - IF, ID, EX, etc....
- Advantages:
  - Removes all WAR and WAW hazards
  - Like Tomasulo, good for allowing full out-of-order completion
  - Allows data to be fetched from a single register file
  - Makes speculative execution/precise interrupts easier:
    - » All that needs to be "undone" for precise break point is to undo the table mappings

Register Renaming Architecture

Explicit Register Renaming

- Make use of a *physical* register file that is larger than number of registers specified by ISA
- Keep a translation table:
  - ISA register => physical register mapping
  - When register is written, replace table entry with new register from freelist.
  - Physical register becomes free when not being used by any instructions in progress.



Questions and Answers

- Solved Examples
  - Loop Unrolling 1
  - Loop Unrolling 2
  - VLIW Eg 1
  - VLIW Eg 2
  - VLIW Eg 3